A Calibrated Cumulative State-Tracking Probe with Initial Predictive Validity for Open-Weight LLM Agent Performance

Dengzhe Hou[1,†] Lingyu Jiang[1] Deng Li[2] Zirui Li[1]
Fangzhou Lin[1,3,4] Kazunori D Yamada[1]

[1]Tohoku University [2]Lappeenranta-Lahti University of Technology LUT
[3]Texas A&M University [4]Worcester Polytechnic Institute

hou.dengzhe.t8@dc.tohoku.ac.jp

[†]Corresponding author.

Abstract

Question. Does a controlled probe of cumulative arithmetic state tracking predict downstream agent performance beyond what task-completion scores reveal? Method. We introduce WMF-AM, a no-scratchpad probe evaluated on 20 open-weight models from 13 families (0.5B–35B) against a released deterministic 10-task agent battery. Main result. In our pre-specified primary analysis (Bonferroni-corrected at α=0.025), WMF-AM achieves Kendall's τ=0.612 (p<0.001, 95% CI $[0.360, 0.814]$, N=20)—within this open-weight sample and battery—consistent in a held-out check (N=21). Three targeted construct-isolation ablations (K=1 control, non-arithmetic ceiling, yoked cancellation) confirm that cumulative accumulation under load—not single-step arithmetic or generic entity tracking—is the primary source of WMF-AM difficulty. Positioning. WMF-AM is not the first arithmetic state-tracking benchmark (Minerva, ICML 2025); it contributes (a) a construct-isolation ablation suite, (b) K-calibration that keeps models in the discriminative range (Minerva QS at K=200 floors all 20 models), and (c) pre-specified probe-to-battery predictive validity, to our knowledge previously undemonstrated for a calibrated process probe. Scope. The sole empirical contribution is the validation of WMF-AM as a predictor of Agent Battery Score in this open-weight sample. Exploratory partial τ analyses are hypothesis-generating only. Generalization to closed API models and broader populations remains open.

1 Introduction

This paper presents a predictive-validity study for a process-sensitive probe. It does not propose replacing task-completion evaluation; it asks whether a controlled probe of cumulative arithmetic state tracking provides incremental predictive value for downstream agent performance beyond completion. Task-completion metrics remain useful and widely applicable. Their limitation, documented here and in prior work, is that completion scores may compress variance in process-level capabilities that matter for reliable multi-step agent behavior—we call this the limits of completion-only evaluation ("Completion Fallacy" shorthand: the inference that completion success alone is sufficient for inferring latent process competence). Benchmarks such as HotpotQA [Yang et al., 2018], MMLU [Hendrycks et al., 2021a], and AgentBench [Liu et al., 2023] measure what LLMs accomplish; process-sensitive probes ask

how they process information under load, as a complementary—not competing—evaluation lens.

Several recent results converge on this point. PAE [Xie et al., 2026] finds 27–78% of agent successes involve corrupted trajectories; Fragile Thoughts [Aravindan and Kejriwal, 2026] and Robust Answers Fragile Logic [Jiang et al., 2026] show correct answers produced despite broken reasoning chains; Turpin et al. [Turpin et al., 2023] demonstrate unfaithful chain-of-thought explanations. These are instances of multiple realizability [Putnam, 1967]: identical outcomes produced by categorically different processes—a pattern also documented as underspecification [D'Amour et al., 2022] and shortcut learning [Geirhos et al., 2020].

This paper introduces and empirically validates WMF-AM, a controlled probe for cumulative arithmetic state tracking under load. The WMF-AM validation is the sole empirical contribution. We situate it within an interpretive taxonomy (CEF) that provides nomenclature and motivation for this and related probes; CEF is explicitly a working conceptual taxonomy, not a validated empirical contribution of this paper. The probe is conceptually related to entity tracking [Kim and Schuster, 2023] and latent state persistence [Anonymous, 2025a]; to state-tracking benchmarks that probe object/entity state (without arithmetic accumulation or ablation controls) [Rezaee et al., 2025, Li et al., 2025]; and to scaling studies of LLM state dynamics [Anonymous, 2025b]. WMF-AM is distinguished by cumulative arithmetic load manipulation, targeted ablation controls (yoked cancellation, K=1, non-arithmetic ceiling), and downstream predictive validity on a deterministic agent battery.

This paper makes four contributions. (1) Measurement construct and probe. We operationalize cumulative arithmetic state tracking under load as a measurable process capability and introduce WMF-AM, a controlled probe for testing it (Section 3). (2) Pre-specified primary predictive validity. Across 20 open-weight models from 13 families, WMF-AM predicts performance on an independent released deterministic battery (τ=0.612, p<0.001, 95% CI $[0.360, 0.814]$; pre-specified, Bonferroni-corrected primary analysis, Section 4). (3) Pre-specified template stability (τ=0.524, p=0.006; pre-specified). Exploratory (hypothesis-generating only): partial τ analyses suggest the predictive relationship survives controls for task completion (partial τ=0.411) and model scale (partial τ=0.503); these require held-out replication and should not be treated as confirmatory evidence. Non-arithmetic ablation, K=1 control, and yoked cancellation confirm the signal is not explained by single-step arithmetic or generic tracking. (4) Robustness. The signal is stable across seeds, templates, paraphrases, and leave-one-out analyses (Section 4). A glossary of all acronyms appears in Table 11 (Appendix F).

Scope. The empirical contribution is the introduction and validation of WMF-AM as one process-sensitive probe of cumulative arithmetic state tracking, evaluated on 20 open-weight models. The two pre-specified confirmatory analyses are predictive validity (τ=0.612) and template stability (τ=0.524); all other results are exploratory. Generalization to closed API models, additional downstream criteria, and broader model families requires held-out evaluation. The broader CEF taxonomy is an interpretive framework, not an empirical result of this paper.

Claim hierarchy. The sole empirical contribution of this paper is the validation of WMF-AM as a predictor of Agent Battery Score in a 20-model open-weight sample. All other analyses either motivate the construct (Sections 2–3), stress-test that primary claim (robustness and ablation results, §4.2), or generate hypotheses for future replication (exploratory partial τ, architectural observations, §4.3). The CEF taxonomy is an interpretive framework, not an empirical result of this paper.

Paper roadmap. Section 2 formalizes the problem and reviews related work; Section 3 introduces WMF-AM and the CEF framework; Section 4 reports the validation study; Section 5 discusses limitations and implications. Table 9 in Appendix E gives development status for all 11 CEF sub-dimensions.

2 Motivation and Definitions

Completion scores may compress variance in process-level capabilities: models with near-identical completion can have sharply divergent WMF-AM scores (within this sample, 30/105 model pairs with $|\Delta\text{completion}| \leq 0.05$ show $|\Delta\text{WMF-AM}| \geq 0.10$). Multiple realizability [Putnam, 1967]—correct outcomes produced via divergent processes—is empirically documented at scale [Aravindan and Kejriwal, 2026, Jiang et al., 2026, Xie et al., 2026, Fan et al., 2026]. Full motivation, three failure modes, empirical support, and positioning relative to related work appear in Appendix A.

Formal definitions. Let $Y_i \in [0,1]$ denote the outcome score and $Z_i \in [0,1]$ a process-sensitive measure for model i. Outcome metrics are statistically insufficient w.r.t. criterion D if partial $\tau(Z, D \mid Y) > 0$ at conventional α (benchmark-conditional; not universal). In this study, partial $\tau(\text{WMF-AM}, \text{Agent} \mid Y) = 0.411$ ($p = 0.011$, N=20; exploratory). Falsification: undermined if process measures never provide incremental prediction beyond completion.

3 WMF-AM and Its Interpretive Framework (CEF)

CEF (Cognitive Evaluation Framework) is a design lens that organizes process-sensitive probes into four proposed dimensions: WMF (Working Memory Fidelity), MCC (Metacognitive Calibration), EMC (Episodic Memory Coherence), and CLA (Cognitive Load Adaptation), with 11 sub-dimensions. This dimensional structure is hypothesized, not validated in this paper (factor analysis requires $N \geq 30$). Full CEF background, design principles, dimension definitions, and cognitive-science anchors appear in Appendix C. Glossary: Table 11 (Appendix F).

Flagship probe: WMF-AM. WMF-AM (Active Manipulation) is a probe of cumulative arithmetic state tracking under load, inspired by working-memory span paradigms. An initial numeric state is presented, K sequential additive/subtractive operations arrive, and the model must report the final cumulative state. The pilot uses depths $K \in \{3, 5, 7\}$ (dissociation study) and $K \in \{2, 4, 6, 8, 12\}$ (yoked cancellation control), across three surface forms (points scoring, warehouse inventory, bank accounts). Prior entity tracking [Kim and Schuster, 2023] tests passive retrieval; WMF-AM tests sequential arithmetic transformation under load. Scope of the construct: WMF-AM specifically measures difficulty with maintaining a running cumulative sum across K additive/subtractive operations. It does not claim to measure general working-memory capacity, and it may partially reflect standalone arithmetic competence; a completed last-operation-only (K=1) control addresses this: 13/15 models achieve near-ceiling single-step arithmetic accuracy ($\bar{x}$=0.957) versus $\bar{x}$=0.151 at K=7, indicating that cumulative accumulation under load—not arithmetic skill per se—is the primary difficulty driver.

Worked example (K=3, points form): "Alice starts with 10 points. Alice gains 5 points. Alice loses 3 points. Alice gains 7 points. What is Alice's current score?" Correct answer: 19. Scoring: exact match (1 if the model outputs 19, 0 otherwise). A model that tracks each update cumulatively ($10 \to 15 \to 12 \to 19$) succeeds; a model that performs only the last operation or guesses a plausible number fails. At K=7, the chain length taxes state maintenance, producing the diagnostic depth×accuracy interaction. Yoked control example (K=3): "Alice starts with 10 points. Alice gains 5 points. Alice loses 5 points. Alice gains 3 points. Alice loses 3 points. Alice gains 7 points. Alice loses 7 points. What is Alice's current score?" Correct answer: 10 (initial state). Each operation is immediately followed by its inverse, requiring identical arithmetic but no cumulative tracking.

Preliminary pilots for MCC, EMC, and CLA are reported in Appendix N–O; construct-validity framework and falsification criteria are in Appendix I.

4 Validation Study: WMF-AM Provides Initial Predictive Evidence for Agent Performance

Variable definitions. Three scores are used throughout this section and should not be confused. WMF-AM Score: the probe output—mean accuracy over $K \in \{3, 5, 7\}$ depths, 4 seeds, 3 surface forms for each model (Section 3). Outcome Correctness (OC): the 100-item general-knowledge completion battery used as a general-capability baseline (Section 4, Phase 1); this is the "completion score" referred to throughout. Agent Battery Score (ABS): the 10-task deterministic downstream criterion (Phase 6); this is the "agent score" or "downstream target" used in all predictive validity analyses. These three scores are distinct: OC and WMF-AM both serve as predictors; ABS is the downstream criterion.

Statistical methodology. All rank correlations use Kendall's τ-b (ties accounted for). Partial τ uses Somers residualization [Somers, 1962]. Bootstrap 95% CIs: 10,000 model-level and family-clustered resamples; family-clustered CI $[0.341, 0.815]$ is virtually identical to model-level $[0.360, 0.814]$—no single family is load-bearing ($\tau \in [0.503, 0.649]$ across leave-family-out runs). Pre-specified primary analyses (Bonferroni-corrected, α=0.025): (1) predictive validity ($\tau = 0.612$) and (2) template stability ($\tau = 0.524$). Both were frozen before expansion-set collection (see PREREGISTRATION.md at the project repository). All other analyses are exploratory (hypothesis-generating only).

Table 1. Study map: complete analysis register. All τ = Kendall's τ-b. C = pre-specified primary (Bonferroni-corrected at α=0.025); E = exploratory (hypothesis-generating only; require held-out replication before drawing strong conclusions). Predictor: WMF-AM Score; Criterion: Agent Battery Score (ABS); Baseline: Outcome Correctness (OC).

Relation	Subset	τ	N	p	95% CI	Type
WMF-AM → Agent overall	All models	0.612	20	<0.001	$[0.360, 0.814]$	C
Bare ↔ Chat template ranks	15 models	0.524	15	0.006	—	C
WMF-AM → Agent (partial | completion)	All models	0.411	20	0.011	—	E
WMF-AM → Agent (partial | log params)	All models	0.503	20	<0.01	—	E
WMF-AM → Agent (partial | compl. + log params)	All models	0.389	20	0.016	—	E
WMF-AM → Agent non-WMF (7 tasks)	All models	0.623	20	<0.001	—	E
WMF-AM → Agent WMF-heavy (3 tasks)	All models	0.527	20	0.003	—	E
WMF-AM ↔ completion	15 models	0.279	15	0.150	$[-0.156, 0.637]$	E
CONV-RFPOC ↔ AGENT-PQ	7 models	0.905	7	0.003	—	E

Note on sample sizes. Different analyses use different subsets: N=20 for all predictive validity and completion-battery analyses (the full study set); N=15 for WMF-AM template stability and divergent validity (original 7 + expansion 8; before small models were added); N=7 for convergent validity only (original models with AGENT-PQ and CONV-RFPOC data collected). All sample sizes are reported explicitly at each result.

Twenty open-weight models from thirteen architectural families form the study set: seven original models (Qwen 7B/14B/32B, Llama 3.1:8b, Gemma 2:27b, DeepSeek-R1:14b, Mistral:7b), eight expansion models (Phi-3:14b, Gemma-2:9b, Qwen-2.5:3b, Llama-3.2:3b, DeepSeek-R1:7b, Mixtral:8x7b, Command-R:35b, Yi:34b), and five small baseline models (0.5B–2B parameters; Table 3).

Models were evaluated in six phases. Phase 1 (Outcome Correctness) used a 100-item battery spanning five domains (factual, math, science, reasoning, language) at five difficulty levels, temperature = 0. Phase 2 (WMF-AM) administered 15 probes at depths $K \in \{3, 5, 7\}$ with 4 independent random seeds for all 20 models. Phase 3 (MCC-MA) applied a 20-problem self-monitoring probe. Phase 4 (Yoked Cancellation Control) administered 20 trials per depth at $K \in \{2, 4, 6, 8, 12\}$ using operations that cancel (gain then lose, transfer then reverse), so the correct answer equals the initial state despite requiring identical arithmetic parsing as WMF-AM. Phase 5 (Template Harmonization) re-administered WMF-AM under three prompt wrappers (bare, chat, chain-of-thought) to test sensitivity to surface format. Phase 6 (Agent Validation) administered 10 multi-step agent tasks (tool use, multi-step reasoning, state tracking) to all 20 models to test predictive validity. All models are Ollama-hosted open-weight variants using default chat templates, fp16 quantization where available,

Table 2. Study design summary. All phases: temperature = 0, greedy decoding, Ollama-hosted models with default chat templates. All 20 models evaluated with 4 seeds for WMF-AM.

Phase	Probe	Depths	Items	Seeds	Scoring
1. Outcome	100-item battery (5 domains)	—	100	1	Exact match (0/1)
2. WMF-AM	State tracking	K=3, 5, 7	15/depth	4	Exact match on final state
3. MCC-MA	Self-monitoring	—	20	1	Pearson r (predicted vs actual)
4. Yoked	Cancellation ops	K=2, 4, 6, 8, 12	20/depth	1	Exact match on initial state
5. Template	3 prompt wrappers	K=3, 5, 7	15/depth	1	Rank stability (τ)
6. Agent	10 multi-step tasks	—	10	1	Process quality (0–1)

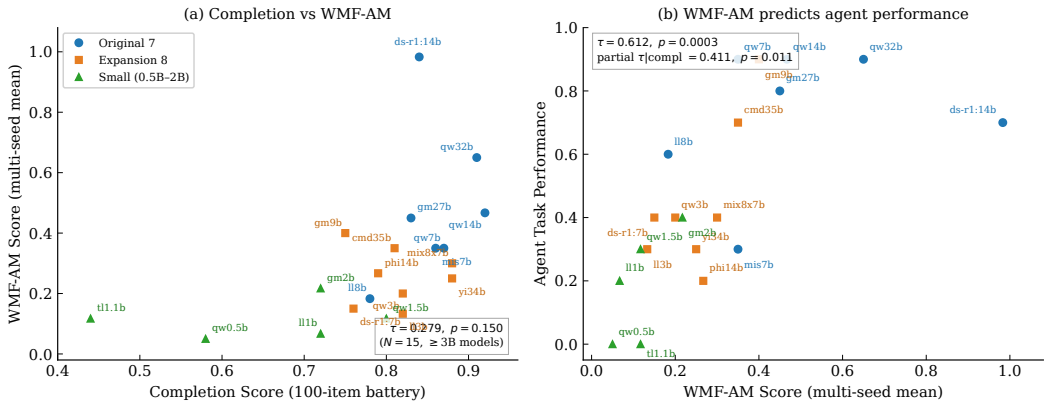

Figure 1. (a) Completion clusters while WMF-AM spans a wide range (N=20, 13 families). (b) WMF-AM predicts downstream agent performance (τ=0.612, p<0.001; pre-specified confirmatory). Exploratory: partial τ=0.411 (p=0.011) after controlling for completion (same sample; requires held-out replication). Blue circles = original 7; orange squares = expansion 8; green triangles = small models (0.5B–2B).

and greedy decoding (temperature = 0, no top-p sampling). No task overlap exists between phases (full study matrix in Appendix K). Table 2 summarizes the experimental design.

4.1 Primary Confirmatory Results

WMF-AM predicts Agent Battery Score beyond Outcome Correctness. Figure 1 and Table 3 present the key finding across N=20 models from 13 families. WMF-AM (4-seed means) spans 0.050–0.983 ($\Delta = 0.933$) while the 100-item completion battery spans 0.44–0.92 ($\Delta = 0.480$). Among the 15 mid-to-large models (≥3B), completion clusters at 0.75–0.92 (SD = 0.052) while WMF-AM spans 0.133–0.983 (SD = 0.228): completion masks substantial process-quality differences. Predictive validity: WMF-AM significantly predicts performance on a deterministic 10-task multi-step agent battery within this 20-model open-weight sample ($\tau = 0.612$, $p < 0.001$, N=20; bootstrap 95% CI $[0.360, 0.814]$). Robustness check — out-of-sample (N=1 held-out model): To provide initial out-of-sample evidence, we ran the full WMF-AM probe and agent battery on llama3.1:70b, a Llama-family model withheld from the 20-model study set. Its WMF-AM score (0.467) placed it 4th of 21 by the probe; its agent score (0.700) placed it 8th of 21 by the criterion — a rank gap of 4, within the concordant range. The τ statistic is essentially unchanged when the held-out model is included: $\tau = 0.627$ ($p = 0.0002$, N=21) vs. $\tau = 0.626$ (N=20). This single held-out point is consistent with the in-sample association but does not constitute a definitive out-of-sample replication; broader held-out validation with API models across diverse families remains future work. Exploratory incremental validity: partial $\tau = 0.411$ ($p = 0.011$) after controlling for completion, and partial $\rho = 0.437$ ($p = 0.008$) via Spearman (robustness check), are consistent across two rank-correlation families; these are exploratory results on the same N=20 sample and should be treated as hypothesis-generating, pending broader replication. Completion also predicts agent performance ($\tau = 0.428$, $p = 0.012$). The primary pre-specified evidence is twofold: (1) significant pre-specified predictive validity ($\tau = 0.612$) and

Model	Family	Outcome	WMF-AM	Agent	Yoked	WMF−Yoked
Original 7 (all 4-seed WMF-AM)						
deepseek-r1:14b	DeepSeek	0.840	0.983	0.70	0.810	+0.173
qwen2.5:32b	Qwen	0.910	0.650	0.90	1.000	−0.350
qwen2.5:14b	Qwen	0.920	0.467	0.90	0.710	−0.243
gemma2:27b	Gemma	0.830	0.450	0.80	0.620	−0.170
qwen2.5:7b	Qwen	0.870	0.350	0.90	0.770	−0.420
mistral:7b	Mistral	0.860	0.350	0.30	0.550	−0.200
llama3.1:8b	Llama	0.780	0.183	0.60	0.340	−0.157
Expansion 8 (all 4-seed WMF-AM)						
gemma2:9b	Gemma	0.750	0.400	0.90	0.960	−0.560
command-r:35b	Cohere	0.810	0.350	0.70	0.920	−0.570
mixtral:8x7b	Mistral	0.880	0.300	0.40	1.000	−0.700
phi3:14b	Phi	0.790	0.267	0.20	0.950	−0.683
yi:34b	Yi	0.880	0.250	0.30	1.000	−0.750
qwen2.5:3b	Qwen	0.820	0.200	0.40	0.970	−0.770
deepseek-r1:7b	DeepSeek	0.760	0.150	0.40	0.940	−0.790
llama3.2:3b	Llama	0.820	0.133	0.30	0.510	−0.377
Small models (broader baseline)						
gemma2:2b	Gemma	0.720	0.217	0.40	—	—
qwen2.5:1.5b	Qwen	0.800	0.117	0.30	—	—
tinyllama:1.1b	TinyLlama	0.440	0.117	0.00	—	—
llama3.2:1b	Llama	0.720	0.067	0.20	—	—
qwen2.5:0.5b	Qwen	0.580	0.050	0.00	—	—
Δ	—	0.480	0.933	0.90	—	—

Table 3. Per-model scores: WMF-AM Score, Outcome Correctness (OC), Agent Battery Score (ABS), and Yoked control accuracy (N=20, 13 families). WMF-AM Score = 4-seed mean at $K \in \{3, 5, 7\}$; OC = 100-item general-knowledge battery (Outcome Correctness; baseline predictor); ABS = 10-task deterministic agent battery (downstream criterion; pre-specified primary analysis). WMF-AM (pre-specified primary): τ=0.612 (p<0.001, confirmatory, C). OC → ABS: τ=0.428 (p=0.012, exploratory, E). Yoked column: cancellation-control accuracy (original 15 models only).

(2) significant pre-specified template stability ($\tau = 0.524$, $p = 0.006$).

4.2 Robustness and Construct Isolation

Completion battery design and broader baseline. Phase 1 uses a 100-item battery spanning five domains (factual, math, science, reasoning, language) at five difficulty levels, yielding score resolution of 0.01. Scope of the comparison: This battery is a custom general-knowledge proxy for overall model capability, not a standard public benchmark suite. It serves as a practical within-study control for general capability; it does not represent completion on the same downstream tasks. Exploratory: the log(parameter count) control ($\tau = 0.503$, $p < 0.01$) provides a complementary model-quality proxy that does not depend on the battery choice, and the joint partial τ ($\tau = 0.389$, $p = 0.016$) controls for both simultaneously (hypothesis-generating; requires held-out replication). With the addition of 5 small models (0.5B–2B), scores now range from 0.44 (tinyllama:1.1b) to 0.92 (qwen2.5:14b) (SD = 0.131, N=20), a substantial improvement over the original 15-model range of 0.75–0.92 (SD = 0.052). The broader range strengthens interpretation: completion now correlates with agent performance ($\tau = 0.428$, $p = 0.012$), but WMF-AM provides significant incremental prediction (partial $\tau = 0.411$, $p = 0.011$). Among the 15 mid-to-large models (≥3B), the outcome range remains compressed (SD = 0.052), confirming that range restriction is a property of the model population, not the battery design. Difficulty-level analysis confirms the battery discriminates (level-5 items: 0.0–1.0 spread; level-1 items: 0.73–0.95).

Exploratory architectural observations (Appendix U). Only deepseek-r1:14b shows positive WMF−Yoked residual (+0.173), suggesting genuine cumulative tracking; all other 14 models show negative residuals, with the largest gaps in smaller models. Figure 3 shows depth-dependent degradation curves (N=15).

Alternative explanations and template sensitivity. WMF-AM scores may reflect nuisance factors beyond state tracking: instruction-following ability, tokenizer effects on numeric parsing, scratchpad/chain-of-thought policy differences, or prompt-format sensitivity. Phase 5 (template harmonization) directly tests the last concern: across 15 models × 3 templates, model rankings under bare and chat wrappers are significantly correlated ($\tau_{\text{bare,chat}} = 0.524$, $p = 0.006$), indicating that WMF-AM rankings are moderately robust to surface-format variation. Chain-of-thought (CoT) templates push nearly all models to ceiling (≥ 0.93), eliminating variance.

Target construct clarification and boundary condition: WMF-AM is designed to measure default state-tracking capability under the no-scratchpad regime (no chain-of-thought, no externalized intermediate steps). It deliberately probes the model's internalized processing. The CoT ceiling effect (≥ 0.93 for nearly all models) is therefore not a validity threat but a measurement boundary: WMF-AM is explicitly a probe of the no-externalization regime, which is the relevant capability for deployment contexts where intermediate outputs are not monitored [Nye et al., 2021]. Implication for generalization: WMF-AM rankings under no-scratchpad conditions predict performance on this battery under the same no-scratchpad evaluation setting; whether WMF-AM scores predict performance under CoT-enabled, tool-augmented, or alternative agent battery regimes is an open empirical question outside the scope of this paper. The moderate bare↔chat stability ($\tau = 0.524$, not near 1.0) indicates some prompt sensitivity remains within the no-scratchpad regime; this is a known limitation. The yoked control isolates arithmetic parsing, but instruction-following and tokenizer effects remain live confounds. Ablation results (N=15): (i) Non-arithmetic state tracking (3 domains × $K \in \{3, 5, 7\}$, 10 problems each): when operations involve direct-assignment updates (color "Alice's color changes to blue", location, status) instead of numeric accumulation, models achieve near-ceiling accuracy (mean 0.98, range 0.91–1.00), versus the WMF-AM arithmetic range of 0.20–0.72. Direct-assignment tracking is solved by remembering the most recent value per entity; arithmetic tracking requires maintaining a running cumulative sum across K additive/subtractive operations—a strictly harder representational demand. The floor–ceiling contrast confirms that WMF-AM difficulty is driven by arithmetic accumulation under load, not generic entity-state tracking. This does not rule out that WMF-AM partially reflects standalone arithmetic skill; a completed last-operation-only (K=1) control addresses this: 13/15 models achieve near-ceiling single-step arithmetic accuracy (mean 0.957) yet floor at K=7 accumulation (mean 0.151), ruling out single-step arithmetic as the primary difficulty driver. Broader multi-step serial-composition alternatives (e.g., error propagation, sequential prompt processing) cannot be excluded without further mechanistic analysis. (ii) Prompt paraphrase (5 templates: original, formal, casual, minimal, verbose): model rankings are stable across natural-language rephrasings; mean cross-template Kendall $\tau = 0.54$ (5/6 pairs $p < 0.05$); original-vs-formal $\tau = 0.857$ ($p < 0.001$). The stripped "minimal" template induces task failure rather than rank reversal, consistent with insufficient-context degradation rather than construct invalidity. Rankings on the paraphrase-original template correlate near-perfectly with the main WMF-AM results: $\tau = 0.951$ ($p = 0.004$, N=7 overlapping models). (iii) Completed in prior loops: three surface forms (points, inventory, accounts), three prompt wrappers (bare, chat, CoT), and four random seeds—all showing stable inter-model rank orderings.

4.3 Predictor Comparison and Exploratory Results

ABS (Agent Battery Score) is the mean of 10 multi-step task scores spanning state-tracking, multi-step reasoning, general problem-solving, and constrained planning—each deterministically verified by exact match; see Appendix S for the full 10-task table and battery release.

Table 4 shows WMF-AM against five predictors. The yoked cancellation control (arith-

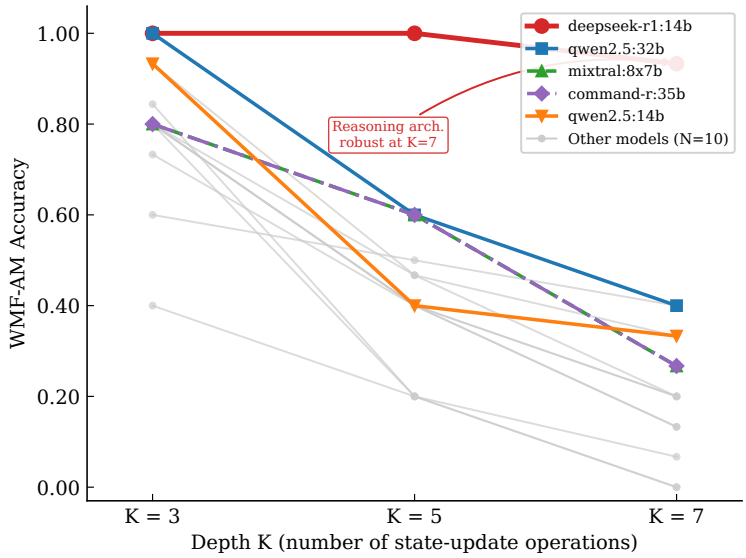

Figure 2. WMF-AM accuracy by depth K (N=15, chat template). deepseek-r1:14b (red) maintains near-perfect accuracy; all others degrade sharply at $K \geq 5$. Top 5 models highlighted.

metic parsing only: τ=−0.031, p=0.879) confirms that the predictive signal is specific to cumulative state-tracking, not arithmetic parsing ($\Delta\tau$=0.491, p=0.022). Exploratory partial τ controlling for completion (0.411, p=0.011) and scale (0.503, p<0.01) are hypothesis-generating; full details in Appendix Q.

Table 4. Predictor comparison: WMF-AM Score vs. alternative predictors of ABS. C = pre-specified; E = exploratory. $\Delta\tau$ (WMF vs. Yoked): 0.491 (p=0.022, bootstrap, inferentially confirmed). $\Delta\tau$ (WMF vs. completion/params): directional but not confirmed (95% CI overlaps zero).

Predictor	τ	p	N	Note
WMF-AM (this paper)	0.612	<0.001	20	C
log(params)	0.444	0.010	20	E
Completion score	0.428	0.012	20	E
K=1 single-step accuracy	0.411	0.047	15	E
Yoked cancellation accuracy	−0.031	0.879	15	E (parsing only)

4.4 Measurement Robustness

Measurement robustness. Multi-seed reliability: all 15 models evaluated with 4 independent seeds × 15 probes. Expansion-model SDs range 0.073–0.144 (mean 0.108), comparable to original-model variability. Cross-template stability: $\tau_{\text{bare,chat}} = 0.524$ ($p = 0.006$); $\tau_{\text{chat,cot}} = 0.314$ ($p = 0.114$). Leave-one-family-out robustness (exploratory): Removing all models from each family in turn, the WMF-AM→Agent τ ranges 0.503–0.649 (mean 0.588). Removing the Qwen family (N=6 removed, N=14 remaining): $\tau = 0.617$; removing DeepSeek (N=3 removed, N=17 remaining): $\tau = 0.570$. The effect is not driven by any single family. Family-aggregated analysis (exploratory): Computing per-family median WMF-AM and Agent scores (13 families) and correlating family-level medians yields $\tau = 0.538$ ($p = 0.030$), confirming the relationship is present at the family level and is not solely an artifact of within-family ordering. See Appendix M for full analyses.

Complementary probes (preliminary; Appendix N–P). Three probes at pilot scale ($N \leq 8$, single seed) serve as preliminary existence checks only. EMC-lite: spread = 0.400, $\tau(\text{WMF-AM}, \text{EMC-lite})$=0.250 (consistent with dimensional independence; not confirmed). MCC-CE-v2: error rates 23–40% (floor effect fixed), but 7/8 models show zero self-flagging—

requiring replication. CLA-DC: provisionally collapses onto WMF-AM (τ=0.905, N=7); pending factor analysis. We draw no multi-dimensional structure claims from these pilots.

4.5 Validity Summary

The complete validity register (Table ??, Appendix Q) shows: divergent validity ($\tau_{\text{WMF-AM,completion}}$=0.279, p=0.150; WMF−Yoked vs outcome: p=0.519); convergent validity (CONV-RFPOC vs AGENT-PQ: τ=0.905, N=7; caveat: LLM judge bias); template stability (τ=0.524, p=0.006, confirmatory). CLA-DC provisionally collapses onto WMF-AM; MCC-MA and EMC-lite remain separable ($\tau \leq 0.250$).

5 Limitations, Alternative Views, and Discussion

Comparison to Minerva and K-calibration. Minerva [Xia et al., 2025] (ICML 2025) includes a Quantity State task with K=200 sequential +/- operations. In a full head-to-head on all 20 study models, Minerva QS at K=200 yields 20/20 models at exactly 0.000—entirely non-discriminative, so $\tau(\text{Minerva QS}, \text{Agent})$ is undefined. WMF-AM's K=3/5/7 calibration targets the discriminative window (scores span $[0.050, 0.983]$, yielding τ=0.612). Discriminability degrades greedily with K: τ=0.619 at K=3, τ=0.509 at K=7, τ undefined at K=200 (full K-sweep table: Appendix B). WMF-AM is not a repackaging of Minerva QS; it adds the construct-isolation ablation suite and K-calibration that restores discriminability.

Key limitations. Limitations are organized by severity into four buckets.

(L1) External validity (primary limitation). Results come from a structured, family-clustered, non-random sample of 20 open-weight models (0.5B–35B, greedy decoding, Ollama variants). Claims do not extend to closed API models, fine-tuned variants, or non-English tasks. The single held-out check (llama3.1:70b; τ unchanged at 0.627 with N=21) is consistent with the in-sample association but does not constitute replication. Priority next step: Extend to $N \geq 30$ including GPT-4o, Claude 3.5, and Gemini Pro; re-run WMF-AM and the same battery. Agent tasks test multi-step capability, not robustness under distribution shift; robustness-predictive validity remains open.

(L2) Construct validity. WMF-AM measures cumulative arithmetic state tracking under load in the no-scratchpad regime; it does not measure general working-memory fidelity or general arithmetic skill (both ruled out by the K=1 control). CoT templates collapse all models to near-ceiling ($\geq$ 0.93): this is a construct boundary, not a failure — WMF-AM probes the no-externalization regime. Rival explanations (instruction-following, tokenizer effects) are mitigated by cross-template stability (τ=0.524) but not eliminated (Appendix J). The K=1 exact-match parser has an artifact for two verbose reasoning models (deepseek-r1:14b/7b); 88/90 of their responses contain the arithmetically correct result, so the conclusion is robust. MCC-CE-v2 (binary protocol) may under-elicit partial metacognitive capacity; results are preliminary.

(L3) Sample size and dependence structure. N=20 supports the two pre-specified primary analyses but not factor analysis; the four-dimension CEF structure is a provisional taxonomy, not a validated empirical model. Family non-independence is substantial (6 Qwen, 3 DeepSeek models); within-family ordering may trivially inflate τ. The within-Qwen robustness check (τ=0.894, p=0.016, N=6) is included but family-balanced sampling is needed for full rigor.

(L4) Comparator scope. The completion proxy is a custom 100-item general-knowledge battery, not a standard benchmark (MMLU, GSM8K). Exploratory partial τ controlling for completion or scale are hypothesis-generating only; replication on a held-out model set is needed before interpreting incremental validity.

What WMF-AM is not. WMF-AM is not a measure of general working-memory capacity (ruled out by K=1 control), general arithmetic skill (ruled out by K=1 control), generic entity tracking (ruled out by non-arithmetic ablation), or arithmetic parsing (ruled out by yoked cancellation). What remains unresolved: instruction-following and tokenizer effects are partially mitigated by cross-template stability but not eliminated; CoT scaffolding

bypasses the bottleneck (construct boundary, not failure); and the mechanism by which cumulative arithmetic difficulty predicts agent performance across diverse task categories is not established.

Scope of contribution. WMF-AM is one concrete validated probe in one specific regime: open-weight models (0.5B–35B), evaluated under greedy decoding without scratchpad, on cumulative arithmetic state tracking. The two pre-specified findings (predictive validity τ=0.612; template stability τ=0.524) are initial in-sample evidence for this slice of model space; they do not constitute a general claim about process-sensitive probes predicting agent performance at large. Other process-sensitive probes and frameworks are welcome; the CEF taxonomy is an organizing research agenda, not an empirical result of this paper.

Implications and agenda. Benchmarks would benefit from including process probes alongside completion: a load-varied probe [Zhang et al., 2024], a metacognitive monitoring component [Lightman et al., 2023], and a difficulty gradient. CEF is a proposed research program; this paper establishes WMF-AM as one validated probe within it. Priority next steps: extend to API models ($N \geq 30$), test with standard benchmarks (MMLU, GSM8K), and confirm whether the predictive validity result replicates across probe families.

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A Full Motivation and Related Work

Three failure modes of completion-only evaluation. (F1) Diagnostic blindness: models with near-identical completion scores can have very different process profiles. (F2) Uninformative failure diagnosis: current benchmarks report that an agent failed, not why—whether the bottleneck is state tracking, metacognitive calibration, or episodic confusion. (F3) Safety blindness: pattern-matching agents (which fail predictably at distribution shift) are indistinguishable from genuine reasoners on standard benchmarks.

Empirical support. Multiple realizability is empirically documented at scale. Fragile Thoughts [Aravindan and Kejriwal, 2026] and Robust Answers Fragile Logic [Jiang et al., 2026] show correct answers produced despite corrupted reasoning. Turpin et al. [Turpin et al., 2023] and Lanham et al. [Lanham et al., 2023] show CoT can be systematically unfaithful. PAE [Xie et al., 2026]: 27–78% of successful completions involve corrupted intermediate actions. AgentProcessBench [Fan et al., 2026]: 8,509-step human annotation shows weaker models inflate correct-step ratios via early termination. Chaduvula et al. [Chaduvula et al., 2026]: state-tracking inconsistency is 2.7× more prevalent in failed agent runs (TAU-bench).

Positioning relative to prior work. Entity Tracking [Kim and Schuster, 2023] is prior art for WMF-AM's design; Latent State Persistence [Anonymous, 2025a] probes sequential latent-state manipulation on hidden integers—a closely related paradigm. Minerva [Xia et al., 2025] (ICML 2025): see Table 5. CEF adapts cognitive science measurement logic [Miller, 1956, Baddeley and Hitch, 1974, Flavell, 1979, Nelson and Narens, 1990, Tulving et al., 1972, Sweller, 1988] for API-level LLM evaluation without claiming LLMs have human cognition. Process supervision [Lightman et al., 2023, Uesato et al., 2022] trains models using step-level labels; CEF measures process quality. Mechanistic interpretability explains why a score is low; CEF identifies that it is low.

Table 5. Comparison to Minerva (ICML 2025) Quantity State. WMF-AM does not claim benchmark superiority; it contributes three additions to the existing arithmetic state-tracking paradigm.

Property	Minerva QS	WMF-AM (this paper)
Arithmetic state-tracking task	✓	✓
Step-count variation	✓(K variable)	✓(K=3, 5, 7)
Calibrated for open-weight discriminability	× (K=200: 20/20 models floor at 0.000)	✓(K=3/5/7: range [0.050, 0.983])
Construct-isolation ablation suite	×	✓(yoked, K=1, non-arith)
Downstream predictive validity	×	✓(τ=0.612, N=20)
Kendall τ with agent performance	undefined (constant)	0.626 (p<0.001)
Multi-seed + multi-template stability	×	✓
Head-to-head on same 20 models	N/A	✓(this paper)

B K-Sweep: Discriminability vs. Step Count

Table 6. K-sweep: discriminability vs. step count. τ=τ(probe, Agent); N = models with per-K data. K=3–7: WMF-AM per-depth means; K=20: 5-model inline test; K=200: full 20-model head-to-head.

K	N	Mean acc	Range	τ	p
3	12	0.512	[0.050, 0.950]	0.619	0.007
5	12	0.296	[0.000, 1.000]	0.545	0.020
7	12	0.192	[0.000, 1.000]	0.509	0.032
20	5	0.180	[0.000, 0.900]	−0.333	0.47
200 (Minerva)	20	0.000	[0.000, 0.000]	undef	undef

C CEF Framework Background

CEF (Cognitive Evaluation Framework) organizes process-sensitive probes into four proposed dimensions: WMF (Working Memory Fidelity, 3 sub-dimensions), MCC (Metacognitive Calibration, 2), EMC (Episodic Memory Coherence, 3), and CLA (Cognitive Load Adaptation, 3). Each dimension has a cognitive science anchor, a documented evaluation gap, and API-level testability. Labels like "working memory" are operational shorthand, not mechanistic claims [] (see Appendix G). Table 7 summarizes the four dimensions.

Other CEF dimensions (background; not validated in this paper). MCC: decomposes metacognition into monitoring accuracy (MCC-MA) and control efficacy (MCC-CE). EMC: tests temporal ordering, cross-episode discrimination, and source attribution under interference.

Table 7. CEF dimensions: target, failure signature, and exclusions. Detailed sub-dimension definitions appear in Appendix H.

Dim	Target	Failure signature	Excluded
WMF	Active maintenance and transformation of state under load	Accuracy drops as concurrent items exceed threshold	Long-context retrieval; knowledge recall
MCC	Self-monitoring of errors (MA) and correction of flagged errors (CE)	Confident-wrong; flagging without correction	General calibration (ECE); cross-model detection
EMC	Temporal ordering, cross-episode discrimination, source attribution	Episode blending; source misattribution	Single-turn retrieval; semantic memory
CLA	Degradation shape and resource scaling under load	Cliff-edge failure; non-recovery	Knowledge difficulty; unrelated complexity

CLA: characterizes the performance-versus-load function shape; CLA-DC provisionally collapses onto WMF-AM (τ=0.905, N=7) pending factor analysis at $N \geq 20$.

Construct validity framework. Following Cronbach and Meehl [1955], Messick [1989], Campbell and Fiske [1959], and Jacobs and Wallach [2021], each dimension requires convergent validity (correlation with independent process-sensitive measures) and divergent validity (low correlation with completion). The study supports divergent validity for WMF-AM (τ = 0.279 with a 100-item completion battery, N=15). CEF would be falsified if all dimensions collapsed onto a single factor indistinguishable from task completion. Appendix I provides full construct-to-probe mapping.

D Full Related Work Table

Table 8. Representative benchmarks and evaluation work, grouped by measurement type and CEF gap addressed.

Work	Measures	CEF gap addressed
MMLU [Hendrycks et al., 2021a], ARC [Clark et al., 2018], HellaSwag [Zellers et al., 2019]	Knowledge recall	No memory structure, no process
HumanEval [Chen et al., 2021], MATH [Hendrycks et al., 2021b], GSM8K [Cobbe et al., 2021]	Output correctness	No reasoning fidelity
HotpotQA [Yang et al., 2018], WebShop [Yao et al., 2022]	Task completion	No metacognition, no memory typ
AgentBench [Liu et al., 2023]	Sequential task success	No within-task process
Reflexion [Shinn et al., 2023]	Post-reflection improvement	Monitoring $\neq$ control (conflated)
PAE [Xie et al., 2026]	Corruption rate	Motivates CEF; no per-dimension measurement
NeuroCognition [Haznitrama et al., 2026]	Cognitive ability (IQ)	Ability $\neq$ process quality; no load variation
Strong Mem. & Weak Ctrl. [de Langis et al., 2025]	Executive function	Isolated tasks; no multi-turn load
Entity Tracking [Kim and Schuster, 2023]	Entity-state	Prior art for WMF-AM design (passive retrieval)
Latent State Persistence [Anonymous, 2025a]	Sequential latent-state manipulation	Closest prior paradigm; no arithmetic load, no agent battery
Minerva [Xia et al., 2025]	Programmable memory (incl. Quantity State: sequential +/-)	Closest arithmetic prior; WMF-AM adds construct-isolation suite + agent-predictive validity
Zhang et al. [Zhang et al., 2024]	WM-style probes $\rightarrow$ reasoning limits	Motivational; no ablation suite, no agent battery
Intell. Degradation [Wang et al., 2026]	Load degradation	Prior art for CLA
HELM [Liang et al., 2023]	Behavioral outcomes	No cognitive processes
Long-context [Bai et al., 2024]	Retrieval from static window	WMF-AM targets active transformation
ECE calibration [Guo et al., 2017]	Confidence scalar	MCC decomposes into monitoring/control

E Development Status and Summary Tables

F Glossary

G On Cognitive-Science Labels

We use labels like "working memory" and "metacognition" as operational shorthand for specific behavioral probes, not as claims about LLM internal mechanisms. "WMF-AM" means: "accuracy on sequential state-tracking tasks degrades with depth K"; it does not mean LLMs have a working memory buffer. "MCC-MA" means: "accuracy of self-error

Table 9. Development status of all 11 CEF sub-dimensions. Status key—A: robust differentiation + multi-seed reliability; B: differentiation shown, limited seeds/models; C: probe redesigned, initial results; D: ceiling/floor effect, needs redesign; E: protocol specified, no pilot data.

Sub-dim	Status	Evidence	Limitation
WMF-AM	A^+	Δ=0.933, 4-seed all 20, 3-template (τ=0.524), yoked control, predictive τ=0.612	All open-weight
WMF-IM	B	spread 0.480, N=5	Single paradigm (hard)
WMF-IR	D	Ceiling (= 1.0 all models)	Needs redesign
MCC-MA	B	4/7 models differentiated	Floor for 3 models
MCC-CE	C	v2: error 23–40%, near-zero flagging (N=8)	Binary protocol
EMC-lite	B	Source-conflict spread 0.400	Single seed
CLA-DC	$B^\dagger$	Correlates τ=0.905 with WMF-AM	†Provisionally collapsed onto WMF
CLA-CR	C	v3 spread 1.171, N=7	v1–v2 failed; 2 seeds
CLA-RA	E	—	Awaits full study
EMC-TO	E	—	Awaits full study
EMC-EI	E	—	Awaits full study

Table 10. CEF at a glance. Each sub-dimension's task type, scoring rule, and pilot status. Tier 1 = novel construct; Tier 2 = novel reframing. Pilot: ✓ = pilot-supported, ∼= preliminary, —= protocol only.

Sub-dim	Task (one-line)	Scoring	Tier	Pilot
WMF-AM	Track K sequential state updates, report final state	Exact match	1	✓
WMF-IM	Recall N facts amid M distractors	Recall accuracy	2	∼
WMF-IR	Recall List-A after List-B interference	Proactive interference δ	2	—
MCC-MA	Predict which of 20 answers are wrong	Pearson r (predicted vs. actual)	1	∼
MCC-CE	Flag uncertain answers, revise them	Correction efficacy	1	∼
EMC-TO	Reorder non-chronological events	Kendall τ vs. ground truth	1	—
EMC-EI	Discriminate similar episodes	Accuracy under similarity	1	—
EMC-SA	Attribute facts to source + episode	Joint attribution accuracy	2	—
CLA-DC	Performance across 5 load levels	Smoothness (SD of Δs)	1	—
CLA-RA	Response elaboration vs. difficulty	Pearson r	1	—
CLA-CR	Easy-task recovery after high-load block	Recovery rate	1	∼

prediction on a 20-item probe"; it does not mean LLMs have phenomenal metacognition. The labels are justified because the probes are adapted from validated human measurement paradigms, and they organize what would otherwise be an unstructured collection of tests.

Two caveats are essential. First, we do not require LLMs to have phenomenal cognition or internal states analogous to human working memory. CEF dimensions are process-sensitive behavioral probes that borrow measurement logic from cognitive science. Second, CEF measurements are behavioral (prompt/response), designed to be differentially sensitive to process quality: high MCC-MA requires accurate self-error prediction; graceful CLA-DC requires proportional rather than catastrophic degradation.

H Full Sub-dimension Definitions

WMF-AM (Active Manipulation, Tier 1). Uses a complex span task: initial state, K sequential modifications, report final state. Depths $K \in \{2, 4, 6, 8, 12\}$, three surface forms. Prior entity tracking [Kim and Schuster, 2023] tests passive retrieval; WMF-AM tests sequential transformation.

WMF-IM (Information Maintenance, Tier 2). Embeds N facts in a passage with M distractors; measures recall accuracy. Extends Wang and Sun [Wang and Sun, 2025] to natural-language multi-turn contexts.

WMF-IR (Interference Resistance, Tier 2). Adapts the List-A/List-B proactive interference paradigm. Currently shows ceiling (= 1.0 all models); needs redesign.

MCC-MA (Monitoring Accuracy, Tier 1). Model predicts which of 20 answers are incorrect; Pearson r measures monitoring accuracy. Unlike Gnosis [Ghasemabadi and Niu, 2025] (requires internals), MCC-MA uses prompting under multi-turn load.

MCC-CE (Control Efficacy, Tier 1). Model flags uncertain answers and revises them. Decomposes into flagging rate, correction efficacy, and false alarm rate.

EMC-TO/EI/SA. Temporal ordering (Kendall τ), cross-episode discrimination under similarity, and joint source-episode attribution. All proposed, no pilot data.

Table 11. Glossary of key terms and acronyms.

Term	Definition
CEF	Cognitive Evaluation Framework (this paper)
WMF-AM/IM/IR	Working Memory Fidelity: Active Manipulation / Information Maintenance / Interference Resistance
MCC-MA/CE	Metacognitive Calibration: Monitoring Accuracy / Control Efficacy
EMC-TO/EI/SA	Episodic Memory Coherence: Temporal Ordering / Episodic Interference / Source Attribution
CLA-DC/RA/CR	Cognitive Load Adaptation: Degradation Curve / Resource Allocation / Capacity Recovery
AGENT-PQ/TC	Agent Process Quality / Task Completion (external evaluation)
CONV-RFPOC	Convergent validity probe: Reasoning Fidelity–Product of Cognition

CLA-DC (Degradation Curve, Tier 1). Smoothness score (SD of successive Δs). Provisionally collapsed onto WMF-AM ($\tau = 0.905$).

CLA-RA/CR. Resource allocation (difficulty–elaboration correlation) and capacity recovery (post-load accuracy recovery). CLA-RA has no pilot data; CLA-CR-v3 spread = 1.171.

Composites (all exploratory). WMF = $0.50\times$AM$+0.30\times$IM$+0.20\times$IR; MCC = $0.55\times$MA$+0.45\times$CE; EMC = $0.40\times$EI$+0.35\times$TO$+0.25\times$SA; CLA = $0.50\times$DC$+0.30\times$RA$+0.20\times$CR. We do not propose an overall CEF composite; per-dimension profiles are the primary unit of analysis.

I Construct-to-Probe Mapping and Falsifiability

Table 12. Construct map for validated probes: manipulated factor, observable signature, rival explanations, and falsification criteria.

Probe	Manipulated factor	Observable signature	Plausible rivals	Expected pattern	Falsified if
WMF-AM	Depth K (number of sequential state updates)	Accuracy $\times$ depth interaction	Instruction following; tokenizer parsing; template familiarity; long-context retrieval	Low τ with completion; monotonic depth decline; $\tau_{\text{bare,chat}} > 0.4$; yoked control $\gg$ WMF-AM	High τ with completion (≥ 0.55); no depth effect; template-specific rankings; control degrades at same rate
MCC-MA	Error-laden vs correct items	Pearson r (predicted vs actual errors)	Calibration shortcuts; always-predict-correct bias	Low τ with WMF-AM; model-specific prediction	Uniform predictions; $\tau \approx 1$ with WMF-AM
EMC-lite	Conflict vs control episodes	Source attribution accuracy under interference	Recency bias; keyword matching	Low τ with WMF-AM; fault $>$ control difficulty	No fault–control difference; $\tau \approx 1$ with WMF-AM

J Rival Explanations

For each probe, plausible alternative explanations remain open. WMF-AM: (a) Instruction following—models may fail at depth because they lose track of instructions. The depth gradient partly addresses this. (b) Long-context retrieval—the control addresses this for 5/7 models. (c) Template artifacts—cross-template stability ($\tau = 0.685$–0.781) mitigates but does not eliminate this. MCC-MA/CE: (a) Calibration shortcuts. (b) Prompt sensitivity. EMC-lite: Recency bias; keyword matching. These are the primary targets for the $N \geq 15$ study.

Table 13. Study matrix: all experiments, models, and sample sizes. †llama3.1:70b added for MCC-CE-v2 only.

Experiment	Metric	N_{models}	Seeds	Trials/model
Outcome Correctness (v2)	Accuracy	20	1	100
WMF-AM (dissociation)	Exact match	20	4	180
Agent Validation	Process quality	20	1	10
Template Harmonization	Rank stability	15	1	135
Yoked Cancellation Control	Exact match	15	1	100
MCC-MA	Pearson r	15	1	20
MCC-CE-v2	Flagging rate	8†	1	50
EMC-lite	Composite	7	1	18
CLA-CR-v3	Recovery rate	7	2	7 blocks
AGENT-PQ/TC	Rubric / binary	7	1	10
CONV-RFPOC	Fidelity	7	1	15

K Study Matrix

L Yoked Cancellation Control Details

The yoked cancellation control uses operation pairs that cancel: "Alice gains 3 points" followed immediately by "Alice loses 3 points." The prompt format, entity names, and depth structure match WMF-AM exactly, so the arithmetic parsing demand is identical. The correct answer equals the initial state, but unlike the original inert control, the model must process each operation to verify cancellation. This eliminates the trivial shortcut of ignoring operations.

M Measurement Robustness

Multi-seed reliability. All 15 models evaluated with 4 independent random seeds × 15 probes. Original-model cross-seed $\tau = 0.798$. Expansion-model SDs: phi3:14b = 0.094, gemma2:9b = 0.144, qwen2.5:3b = 0.122, llama3.2:3b = 0.122, deepseek-r1:7b = 0.114, mixtral:8x7b = 0.116, command-r:35b = 0.100, yi:34b = 0.084 (mean SD = 0.108).

Cross-template stability. Phase 5 administered WMF-AM under three prompt wrappers (bare, chat, chain-of-thought) across all 15 models. Bare ↔ chat: $\tau = 0.524$ ($p = 0.006$)—significant rank preservation. Bare ↔ CoT: $\tau = 0.105$ ($p = 0.627$)—CoT pushes all models to ceiling (≥ 0.93), eliminating variance. Chat ↔ CoT: $\tau = 0.314$ ($p = 0.114$). The CoT ceiling effect is interpretively meaningful: externalized reasoning via scratchpad bypasses the working-memory bottleneck that WMF-AM is designed to measure.

Non-ceiling completion proxies. τ(WMF-AM, MMLU) = 0.356 ($p = 0.317$; range 0.25–0.72); τ(WMF-AM, GSM8K) = 0.150 ($p = 0.645$; range 0.05–0.83). Scores from published model cards, not re-evaluated under pilot setup.

Depth profile.

N EMC-lite Results

EMC-lite provides preliminary support for the source-conflict slice. It does not establish EMC as a whole; single-seed evidence only. τ(WMF-AM, EMC-lite) = 0.250: substantial independence.

O MCC-CE-v2 Results

The v1 floor was an artifact of easy questions; v2 models make many errors but show near-zero self-flagging under the binary protocol. Alternative designs (continuous confidence, forced ranking) might reveal partial capacity.

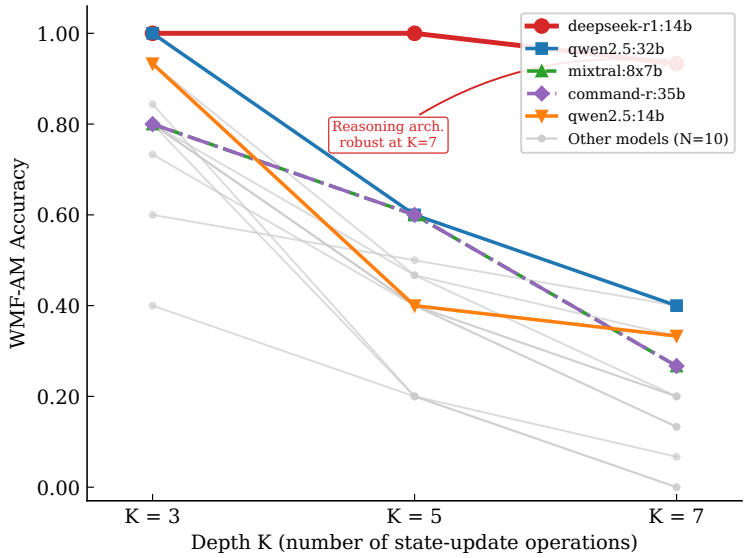

Figure 3. WMF-AM accuracy by depth K (multi-seed means, N=7). deepseek-r1:14b maintains near-perfect accuracy; others degrade at K=7.

Model	F-Det	F-Src	F-Rec	F-Comp	C-Rec	ΔRec
qwen2.5:14b	0.917	0.750	0.917	0.917	0.972	−0.056
qwen2.5:32b	0.917	0.750	0.917	0.917	1.000	−0.083
deepseek-r1:14b	0.889	0.917	0.917	0.912	1.000	−0.083
llama3.1:8b	0.778	0.583	0.750	0.771	0.889	−0.139
qwen2.5:7b	0.861	0.583	0.639	0.736	0.722	−0.083
mistral:7b	0.750	0.417	0.639	0.685	0.722	−0.083
gemma2:27b	0.861	0.583	0.528	0.667	0.611	−0.083
Δ	0.167	0.500	0.389	0.250	0.389	—

Table 14. EMC-lite (N=7, hardened protocol). Source attribution shows strongest differentiation (spread = 0.500).

P Negative OOD Result

Two OOD stress-test designs (v1: 5-step NL tasks; v2: K=8–9 interference tasks) produced anti-correlated CEF–OOD relationships across model pairs. We make no predictive claim. Full assessment requires $N \geq 11$ with process-sensitive downstream tasks.

Q Full Validity Analyses

The validity summary table appears in the main text (Table ??). Below we provide additional details.

Convergent validity details. AGENT-PQ: 10 multi-step scenarios scored by LLM judge (GPT-4o) on 4 rubric dimensions. CONV-RFPOC: 15 reasoning problems with counterfactual chain perturbation. RFPOC $= 1 - P(\text{same answer} \mid \text{perturbed chain})$. Full protocol in Appendix R. Limitation: LLM judge may introduce shared bias.

Construct separation (WMF-AM ↔ CLA-DC). $\tau = 0.905$: no evidence of separability at N=7. Both involve state tracking under load. Planned discriminant test at $N \geq 15$. CLA-CR-v3 provides a structurally independent CLA measure. Even if CLA-DC merges with WMF-AM, WMF-AM, MCC-MA, and EMC-lite remain separable ($\tau \leq 0.250$).

Sensitivity. LOMO for CONV-RFPOC vs AGENT-PQ: τ ranges 0.600–1.000 (mean 0.833). Leave-one-family-out (Qwen, N=4): $\tau = 0.667$. After Holm correction, RFPOC–AGENT-PQ remains significant ($p_{\text{adj}} = 0.021$).

Table 15. MCC-CE-v2 (N=8). Error rates 23–40% (floor eliminated); flagging near zero for all models.

	Balanced (n=30)		Trap-only (n=20)	
Model	Error%	Flag%	Error%	Flag%
qwen2.5:7b	26.7	0.0	40.0	0.0
qwen2.5:14b	36.7	0.0	40.0	0.0
qwen2.5:32b	30.0	0.0	25.0	0.0
llama3.1:8b	26.7	0.0	40.0	0.0
llama3.1:70b	—	—	25.0	0.0
gemma2:27b	30.0	11.1	30.0	0.0
deepseek-r1:14b	23.3	0.0	35.0	0.0
mistral:7b	30.0	0.0	40.0	0.0

Multiple testing. Three primary hypotheses; all others exploratory.

Benchmark governance. All headline results use frozen protocol versions. Probe revisions were motivated by floor/ceiling effects, not model-specific tuning.

R AGENT-PQ Evaluation Protocol

AGENT-PQ uses 10 multi-step scenarios where each model operates within a tool-use scaffold. A separate LLM judge (GPT-4o, temperature = 0) scores each trajectory on four rubric dimensions: plan coherence, state tracking, error recovery, goal adherence. AGENT-PQ = mean of 4 rubric scores/4. AGENT-TC is the binary success rate on the same scenarios.

CONV-RFPOC administers 15 reasoning problems with counterfactual chain perturbation. RFPOC $= 1 - P(\text{same answer} \mid \text{perturbed chain})$.

S Complete Experimental Protocols

Full prompt templates, parameter specifications, scoring procedures, and reference implementations for all 11 sub-dimensions are provided in the supplementary materials.

T Weight Sensitivity Analysis

Three weighting schemes (proposed, equal, PCA-derived) are evaluated. Per-dimension profiles are primary; composites are exploratory only.

U Dissociation Case Studies

Matched dissociation pairs: deepseek-r1:14b and qwen2.5:14b share similar EMC-lite (0.912 vs. 0.917) but differ on WMF-AM (0.983 vs. 0.467). mistral:7b and llama3.1:8b have similar WMF-AM but differ on EMC-lite. These cross-dimension pairs anchor the multiple realizability argument.

V Worked Example: WMF-AM at Depth K=3

Setup. "A warehouse has 3 items: Widget=50, Gadget=30, Sprocket=20." Three updates: "Ship 10 Widgets," "Receive 25 Gadgets," "Transfer 5 Sprockets to overflow."
Correct. Widget=40, Gadget=55, Sprocket=15.
Scoring. Exact match per item. deepseek-r1:14b scores 1.000 at K=7; llama3.1:8b scores 0.050. Both score 0.90–0.95 on outcome correctness. This is the Completion Fallacy in action.

W Novelty Classification

Table 16. Novelty tier for all 11 sub-dimensions. Tier 1 = novel; Tier 2 = novel reframing.

Sub-dim	Tier	Prior art / differentiation
WMF-AM	1	Entity Tracking [Kim and Schuster, 2023]: passive retrieval; Latent State Persistence [Anonymous, 2025a]: structural ops, no arithmetic load, no agent battery; WMF-AM = cumulative arithmetic transformation + ablation suite + downstream validity
WMF-IM	2	Wang & Sun [Wang and Sun, 2025]: key-value pairs; WMF-IM = NL multi-turn
WMF-IR	2	Extended to domain-varied materials
MCC-MA	1	Gnosis [Ghasemabadi and Niu, 2025] requires internals; MCC-MA = prompting only
MCC-CE	1	Reflexion [Shinn et al., 2023]: aggregate; MCC-CE decomposes detection/correction
EMC-TO	1	No prior LLM temporal ordering measurement
EMC-EI	1	No prior cross-episode interference diagnostic
EMC-SA	2	Interference-conditioned attribution
CLA-DC	1	Intell. Degradation [Wang et al., 2026]: magnitude; CLA-DC = shape
CLA-RA	1	Not previously measured
CLA-CR	1	Not previously measured